3.4. VACUUM TUBES

Early days of electronics made successful strides by the introduction and working efficiency of the vacuum tubes. During 20th century, a new branch of engineering called "electronics" originated from the electrical engineering, due to the arrival of the vacuum tubes. These tubes have been finding wide applications in radio, television, long distance telephones, sound motion pictures, radar and electronic computers. A typical vacuum tube having three electrodes called triode is shown in Figure 3.11

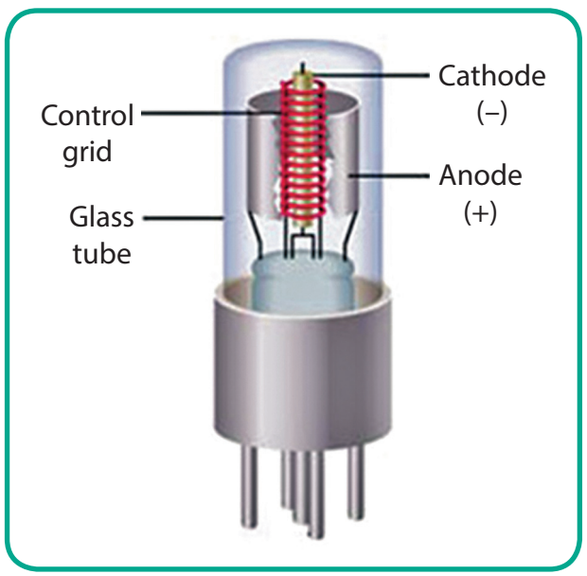


Figure 3.11 Parts of the vacuum tube

Due to its size, slowness in working-speed, cost of production, and above all the emission of heat while on working reduced the life of many electronic instruments.

Continuous research was going on, which paves the way for arrival of semiconductors.

Before studying about semiconductor, it would be better to know about the structure of atom and characteristics of electrons.

3.5 ATOMIC MODEL

The study of atomic structure is very important for electronics engineering. The size of an atom is so small that it is virtually impossible to see it even through the most powerful microscope. Therefore, we have to employ indirect method for the study of its structure. Though many scientists derived atomic theories, Bohr's atomic model is adequate to understand the electronics.

3.5.1. Bohr's Atomic Model

In 1913, Neil Bohr, Danish Physicist gave clear explanation of atomic structure. Bohr postulated the following points about the structure of the atom:



Neil Bohr
(1885 – 1962)

1. An atom consists of positively charged nucleus around which negatively charged electrons revolve in different circular orbits.
2. The electrons can revolve around the nucleus only in certain permitted orbits i.e., orbits of certain radii are allowed.
3. The electrons in each permitted orbit have a certain fixed amount of energy. The larger the orbit (i.e., larger radius), the greater is the energy of electrons.
4. If an electron is given additional energy (e.g., heat, light, etc.), it is lifted to the higher orbit. The atom is said to be in a state of excitation. This state does not lost long, because the electron soon falls back to the original lower orbit. As it falls, it gives back the acquired energy in the form of heat, light or other radiations.

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